Direct Current Subtransmission Backbone for AI Factories:: A Position Paper with Source-Anchored Modeling and Public Benchmark Screens

Author Name(s) Here

Abstract

AI factories are becoming power-delivery-limited before they become compute-limited. The conventional response has been to improve the last portion of the power path, especially by introducing higher-voltage facility buses such as 800 VDC. That direction is necessary, but it is incomplete if alternating current still penetrates the campus deeply enough to recreate repeated conversion, distributed power-quality ownership, and reactive-power coordination across multiple pods or facility front ends. This paper takes the position that the next architectural step is to move the main AC/DC boundary upstream to the campus subtransmission interface and to treat the downstream campus as a DC-native system. To support that position, this paper combines a source-anchored comparative architecture model with OpenDSS AC-boundary consequence screens, a standardized harmonic-sensitivity scan, public RTS-GMLC and SMART-DS benchmark layers, a public ANDES RMS dynamic benchmark, and a reduced-order MVDC fault-envelope screen. Three base scenarios are compared at a 100 MW delivered IT reference campus: Scenario 1, a traditional AC-centric path; Scenario 2, an AC-fed SST plus 800 VDC baseline; and Scenario 3, a proposed 69 kV-class medium-voltage direct-current (MVDC) backbone feeding isolated DC pods and an 800 VDC facility bus. A second model layer then compares Scenario 1(M), Scenario 2(M), and Scenario 3(M) on the same four-block campus topology. In the current model, the MVDC backbone remains favorable in both the single-path and multi-node comparisons relative to that SST baseline. The strongest present evidence still emerges in the multi-node campus and public benchmark screens, where Scenario 3(M) shows consistently lower AC-boundary harmonic sensitivity, lower AC-boundary voltage sensitivity, stronger benchmark voltage robustness under weak remote-corridor stress, and persistence of those advantages under mirrored benchmark locations and local single-branch-outage screens. The paper does not claim that the present model closes all deployment questions. Protection, grounding, harmonic compliance, converter-control dynamics, and proxy-parameter uncertainty still require deeper study. The reduced-order fault screen bounds interruption timescales, but it does not establish a deployable protection design. The position advanced here is therefore narrower than a blanket superiority claim: once AI factories are recognized as large, converter-rich, dynamic loads, a campus-scale MVDC backbone becomes a credible architecture for centralizing AC-boundary power-quality management, reducing upstream voltage sensitivity, and containing fast disturbances at a centralized campus interface, and it merits deeper validation on that basis.

Introduction

AI data centers are entering a regime in which electrical infrastructure increasingly determines scalability. Public demand forecasts already place data centers among the fastest-growing large loads on the grid, while recent AI-factory designs concentrate that demand into highly power-dense campuses [1, 2, 3]. At the same time, public and utility-facing studies now show that AI-training facilities behave as dynamic electrical loads rather than purely static loads [4, 5, 6]. These shifts mean that power architecture can no longer be framed as a rack-level efficiency detail alone.

The conventional campus path remains largely AC-centric. Power enters through AC subtransmission, continues through AC primary distribution, and then encounters a cascade of transformation and conversion stages before it reaches the GPU voltage-regulation stages. Even when modern facility designs adopt 800 VDC internally, AC often remains present up to the facility or pod boundary [7]. That approach improves the last segment of the power

path, but it leaves the campus-level architecture partially unresolved.

This paper advances a position: the correct next move is to establish a DC subtransmission backbone. In that architecture, a common subtransmission AC/DC front end is placed at the campus entry. Downstream of that point, the campus backbone is DC-native, isolation is provided by isolated DC pods or DC transformers, and the facility side remains consistent with the emerging 800 VDC direction. The argument is not that every implementation detail has been solved. The argument is that this architecture is the right reference point for solving the remaining details.

Position

The central claim is that AI-factory power architecture should stop at neither the low-voltage bus nor the pod front end. If the load is increasingly DC-native and the dominant controllable resources such as batteries and photovoltaics are also DC-native, then extending AC deeply into the campus creates avoidable architectural friction.

This position rests on four propositions.

First, repeated high-power conversion stages create a multiplicative efficiency penalty. Optimizing only one stage cannot recover the architectural losses introduced by keeping several large stages in series [8, 9, 10].

Second, AI factories are dynamic loads. Public studies now describe periodic and grid-relevant power fluctuations in large AI-training sites, while utility measurements show that data-center-rich regions can exhibit nontrivial oscillatory behavior linked to power-electronic loads [4, 5, 6]. A campus architecture must therefore be judged by its dynamic interaction with the grid as well as by its steady-state efficiency.

Third, power-quality ownership becomes easier to define when the AC side is centralized. If the campus has one common AC/DC front end, then harmonics and power-factor behavior are concentrated at the grid interface rather than recreated at many downstream AC front ends.

Fourth, a DC backbone is a more natural integration domain for batteries, photovoltaics, and local energy buffers. That does not eliminate the need for sophisticated controls, but it removes unnecessary AC/DC/AC detours from the resource path. In the present paper, the strongest support for this position is at campus scale, where the architecture can be evaluated as a shared backbone across multiple data-center blocks rather than as a single equivalent path.

Scenarios Under Comparison

This paper compares three scenarios derived from the supporting model files in the project workspace.

Three scenarios used throughout this paper. Scenario 3 moves the main AC/DC boundary upstream and keeps the campus DC-native downstream of the centralized front end.

Scenario 1: Traditional AC-centric baseline

The baseline keeps AC as the dominant campus-distribution domain. The representative path is a 69 kV AC campus feeder, medium-voltage to low-voltage transformation, facility UPS, low-voltage AC distribution, a PDU or transformer stage, server AC/DC conversion, and board-level DC/DC regulation.

Scenario 2: AC-fed SST plus 800 VDC baseline

Scenario 2 is intended to represent the advanced AC-fed baseline for this study. In the current scenario, a 69 kV AC campus feeder supplies an explicit SST-style chain: a grid-side rectifier stage, an isolated high-frequency stage, and then an 800 VDC facility bus followed by rack or node DC/DC conversion and board-level DC/DC conversion. The 800 VDC downstream structure is source-backed; the 69 kV campus-side voltage is a local scenario assumption requested for this study [7]. This is the baseline against which Scenario 3 is judged. The intended architectural difference between Scenario 2 and Scenario 3 is therefore the MVDC subtransmission backbone and the upstream relocation of the main AC/DC boundary, not the mere presence of 800 VDC downstream.

Scenario 3: Proposed MVDC backbone

Scenario 3 centralizes the AC/DC boundary at the campus entry. A centralized 69 kV-class AC/DC front end feeds an MVDC backbone, which then supplies isolated DC pods or DC transformers, the 800 VDC facility bus, rack or node DC/DC conversion, and board-level DC/DC regulation. This is the main architectural contribution of the paper. The key system-level change is not merely the use of better converters; it is the relocation of the main AC/DC boundary and the explicit introduction of the MVDC backbone as the campus organizing layer while preserving the same end-use electrical boundary as Scenario 2.

Modeling Framework

Source-anchored architecture model

The first model layer is an architecture comparison model implemented in Python. It evaluates each path from the delivered IT load back to the upstream source by traversing a sequence of stages and conductors. The path efficiency is computed by combining stage efficiencies with conductor losses:

$$\eta_{path} = P_{IT} / P_{source}$$

Annualized electrical loss is then computed over a load-bin set as

$$E_{loss} = \sum_{b \in \mathcal{B}} \left(P_{source,b} - P_{IT,b}\right) H_b,$$

where H_b is the annualized duration of bin b. Partial-load converter efficiencies are represented with source-anchored or explicitly labeled proxy curves. The present assumptions are stored in a source-tagged JSON file so that each major input can be traced to a public source, a local scenario assumption, or a documented proxy.

Dynamic-load layer

The analytical model also includes dynamic AI-load cases. Two frequency anchors are used. The first is a 1.0 Hz reference point within the 0.1--5 Hz band that PNNL identifies as relevant for positive-sequence dynamic studies [4]. The second is a measured 14.8 Hz case drawn from a real data-center oscillation study in Dominion Energy territory [6]. Amplitude values are sensitivity cases rather than universal measured constants. For each case, the model estimates the raw PCC power swing and the native buffer power and energy that would be required at the architecture's natural support point.

For the multi-node campus cases, the dynamic screen also records two explicit metrics:

$$\Delta P_{src,max} = \max \left(\left|P_{src}^{hi} - P_{src}^{0}\right|, \left|P_{src}^{0} - P_{src}^{lo}\right| \right),$$

$$\Delta I_{seg,max} = \max_{\ell \in \mathcal{L}} \left(\left|I_{\ell}^{hi} - I_{\ell}^{0}\right|, \left|I_{\ell}^{0} - I_{\ell}^{lo}\right| \right),$$

where the first is a grid-facing source-swing metric and the second is an internal backbone-current-redistribution metric. This distinction becomes important in the discussion because an architecture can improve the upstream AC boundary while still concentrating dynamic stress internally.

OpenDSS cross-check layer

The second model layer uses OpenDSS as an AC-side quasi-static cross-check tool [11]. This layer is intentionally narrow. It does not attempt to model the internal MVDC backbone or the 800 VDC plant in electromagnetic-transient detail. Instead, it models the upstream AC source, the AC feeder or substation-side AC stub, and the equivalent downstream demand seen at that AC boundary.

For Scenarios 1 and 2, the first modeled 69 kV AC feeder is preserved in OpenDSS. For Scenario 3, the long AC feeder inside the campus is replaced by a short centralized-front-end AC stub because the AC/DC conversion is assumed to occur at the campus entry. This is a surrogate representation of the scenario boundary shift, not a claim of an already-standardized utility model.

The OpenDSS study is executed as a sequence of quasi-static snapshots over two cycles with 64 samples per cycle. At each time step, the downstream demand is updated from the analytical model and the resulting source power, feeder current, feeder loss, and PCC voltage are recorded.

The associated harmonic-sensitivity screen is also intentionally comparative rather than compliance-grade. A standardized per-interface harmonic-current probe is applied at harmonic orders $\mathcal{H}=\{5,7,11,13\}$, and the resulting voltage-distortion proxy is reported as

$$THD_{v,proxy} = \sqrt{\sum_{h \in \mathcal{H}} \left|\Delta V_{POI,h}\right|^2} .$$

This metric is used only to compare scenario sensitivity under the same perturbation basis. It is not an IEEE 519 compliance margin.

Multi-node campus extension

The single-path architecture comparison is useful for isolating end-to-end conversion structure, but it is not by itself a campus-topology model. To address that limitation, a separate multi-node comparison is also run on a shared four-block campus layout. In that extension, Scenario 1(M), Scenario 2(M), and Scenario 3(M) use the same block sizes, tap locations, and inter-block segment lengths. Scenario 1(M) and Scenario 2(M) retain a 69 kV AC backbone, while Scenario 3(M) replaces that campus-interior AC backbone with a 69 kV DC backbone. Each block then carries its scenario-specific local conversion path to the IT load.

This extension is intentionally still simplified: it is radial, balanced, and quasi-static, and it does not yet represent protection or converter-control transients. Its role is narrower. It checks whether the ranking observed in the simpler single-path model survives when the scenarios are forced onto the same explicit multi-block campus topology. A separate public benchmark layer, described next, is then used to test whether the AC-boundary conclusions survive mirrored benchmark locations, local single-branch-outage screens, and non-coherent block-level dynamic patterns.

Public benchmark robustness layer

To avoid relying only on locally defined reference cases, the paper adds a public-data-only robustness layer. Annualized part-load sensitivity is derived from the public NREL ESIF facility IT-power series. Grid-facing voltage and branch-loading sensitivity are screened on the public RTS-GMLC network, and a same-feeder common-network screen is run on a public SMART-DS feeder benchmark [12, 13]. For a given point of interconnection set $\mathcal{P}$, the maximum POI voltage-drop proxy is

$$\Delta V_{POI,max} = \max_{k \in \mathcal{P}} \left(|V_k^{(0)}| - |V_k| \right) \times 100,$$

expressed in percentage points relative to the published RTS operating point. Local N--1 robustness is then summarized as

$$M_{N-1} = \max_{o \in \Omega} M(o),$$

where Ω is the set of one-at-a-time local branch outages around the candidate campus buses, and M is any screened metric such as incremental loading, voltage-drop proxy, or harmonic-sensitivity proxy.

This public benchmark layer is still not a site-specific interconnection study. Its role is to test whether the claimed advantages remain visible when the same scenarios are placed on common public benchmark networks and subjected to mirrored area placements, local branch outages, and more realistic multi-block swing patterns.

Public RMS dynamic benchmark

The public benchmark stack is extended with an electromechanical RMS dynamic study on the public IEEE 14-bus full dynamic case distributed with ANDES [14]. Scenario 2(M) is represented as four distributed AC-fed block interfaces, while Scenario 3(M) is represented as one centralized upstream interface. Disturbance magnitudes are taken from the public MIT Supercloud positive-event burst library and are applied as benchmark load steps rather than as converter-aware controls [15]. The study reports two primary metrics:

$$V_{campus,min} = \min_{k \in \mathcal{P}, t \in \mathcal{T}} |V_k(t)|,$$

$$\Delta f_{campus,max} = \max_{k \in \mathcal{P}, t \in \mathcal{T}} \left| f_k(t) - f_0 \right|,$$

where $\mathcal{P}$ is the set of campus buses, $\mathcal{T}$ is the simulation horizon, and f_0 is the nominal system frequency. A second benchmark mode weakens the remote AC corridor by tripping one published transmission line before the MIT-derived burst step. This remains an RMS benchmark rather than an EMT study, but it strengthens the dynamic evidence at the upstream AC boundary.

Reduced-order fault-duty screen

The paper also adds a reduced-order MVDC fault screen for Scenario 3(M). Starting from the full-load backbone currents of the multi-node campus model, representative backbone and branch locations are evaluated with a simple RL current-rise model,

$$i_f(t) = I_{infty} - left(I_{infty} - I_0right)e^{-t/tau},$$

with

$$I_{infty} = fracV_{dcR_{tot}}, qquad tau = fracL_{totR_{tot}}.$$

Public benchmark values are used for line inductance and breaker-timescale categories [16, 17]. The role of this screen is not to prove a selective protection design. Its role is to bound the interruption timescales and fault-current growth rates that any practical MVDC protection concept would need to address.

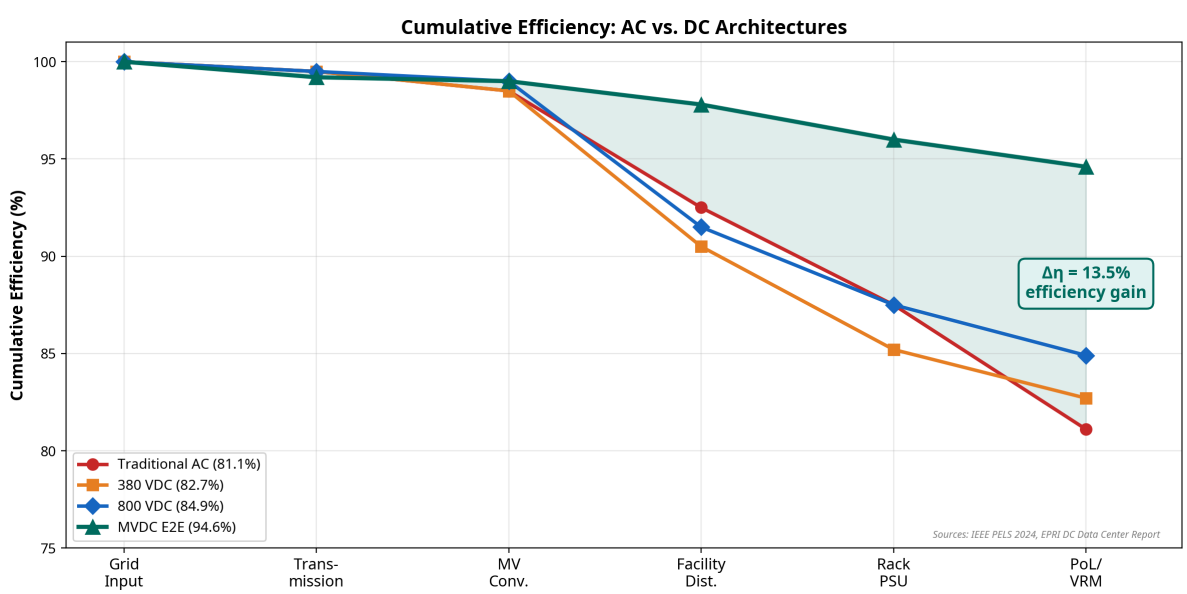
Source basis and limitations

The strongest public anchors in this study come from the IEA, EIA, PNNL, official NVIDIA materials, Open Compute Project data, Schneider Electric specifications, public benchmark repositories, the MIT Supercloud dataset, ANDES, and peer-reviewed literature on DC power conversion, data-center dynamics, and HVDC protection [1, 18, 3, 7, 4, 19, 20, 10, 6, 15, 14, 12, 13, 17, 16]. Some key quantities remain proxy-based, especially the exact partial-load curves for the centralized medium-voltage rectifier and the isolated DC pod stage. Those proxies are labeled accordingly in the model artifacts, and the resulting ranking should be interpreted as conditional on those assumptions rather than as a finalized deployment result.

Results

Steady-state architecture results

Figure fig:efficiency shows the cumulative-efficiency thesis graphically, while Table tab:steady gives the current quantitative results for the 100 MW reference campus.



Cumulative-efficiency comparison for the three evaluated scenarios. The values shown are taken from the current source-anchored model outputs.

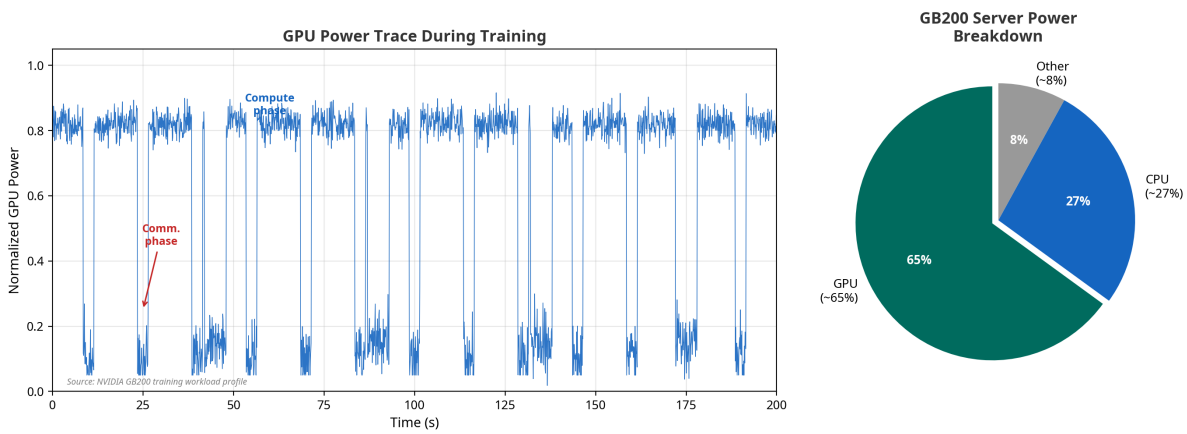
Steady-state comparison at 100 MW delivered IT load

Scenario	Full-load eff.	Input MW	Loss GWh/yr	Loss cost (\$M/yr)
Scenario 1: Traditional AC-centric	87.48%	114.31	125.36	11.65
Scenario 2: AC-fed SST plus 800 VDC baseline	91.82%	108.91	78.06	7.25
Scenario 3: Proposed MVDC backbone	92.17%	108.50	74.45	6.92

The steady-state result is materially traceable and still architecturally relevant after correcting the scenario endpoint. Scenario 3 remains the numerically best case in this reference profile, but the margin over Scenario 2 is still modest: about 0.35 percentage points in full-load efficiency and about 3.61 GWh in annualized losses under the reference continuous-operation profile. That smaller margin matters. It means the strongest remaining argument for Scenario 3 is not a dramatic single-path efficiency gain alone. It is a combination of modest directional efficiency benefit, centralized AC interaction, and a cleaner campus-scale DC architecture.

Dynamic-load results

The need for a dynamic model is motivated by the public evidence that AI-training facilities can exhibit periodic power swings and grid-relevant interactions. Figure fig:gpu summarizes the dynamic-load motivation, and Table tab:dynamic presents selected dynamic results from the analytical model.



Motivation figure emphasizing the dynamic power profile of AI training and the dominance of GPU power in advanced AI systems.

Selected analytical dynamic-load results

Scenario	Case	Raw PCC peak (MW)	Native buffer (MW)	Native buffer (kWh)
Scenario 1: Traditional AC-centric	1.0 Hz, 15% IT amplitude	17.09	15.20	0.67
Scenario 2: AC-fed SST plus 800 VDC baseline	1.0 Hz, 15% IT amplitude	16.32	15.40	0.68
Scenario 3: Proposed MVDC backbone	1.0 Hz, 15% IT amplitude	16.26	15.47	0.68
Scenario 1: Traditional AC-centric	14.8 Hz, 10% IT amplitude	11.35	10.14	0.03
Scenario 2: AC-fed SST plus 800 VDC baseline	14.8 Hz, 10% IT amplitude	10.85	10.27	0.03
Scenario 3: Proposed MVDC backbone	14.8 Hz, 10% IT amplitude	10.80	10.31	0.03

Two points are important. First, the scenario ranking remains only slightly favorable to Scenario 3 when the load is treated dynamically rather than only statically. Second, the margin between scenarios is smaller in the dynamic power-swing metric than in the annual energy metric, which is exactly why dynamic modeling is required. A paper that only reported annual losses would miss an essential part of the AI-factory problem. The correct interpretation

is therefore not that Scenario 3 is proven by this layer. It is only that the backbone architecture remains plausible when the comparison is widened beyond steady-state efficiency. The stronger dynamic evidence later in the paper comes from the multi-node dynamic-diversity screen, which distinguishes grid-facing source swing from internal backbone-current redistribution.

OpenDSS AC-boundary cross-check

The OpenDSS results add a feeder-side cross-check perspective. Table tab:opendssbase reports the 100 MW base-case AC-boundary results, and Table tab:opendssstress reports selected dynamic stress results.

OpenDSS base-case AC-boundary cross-check

Scenario	Source MW	PCC min pu	Feeder loss (MW)	Peak current (A)
Scenario 1: Traditional AC-centric	114.48	0.9707	0.194	1005
Scenario 2: AC-fed SST plus 800 VDC baseline	109.05	0.9722	0.176	956
Scenario 3: Proposed MVDC backbone	108.49	0.9742	0.004	951

OpenDSS dynamic stress results at the AC boundary

Scenario	Case	Source peak (MW)	Max PCC swing	Peak feeder loss (MW)
Scenario 1: Traditional AC-centric	1.0 Hz, 15% IT amplitude	17.15	0.50%	0.259
Scenario 2: AC-fed SST plus 800 VDC baseline	1.0 Hz, 15% IT amplitude	16.40	0.47%	0.234
Scenario 3: Proposed MVDC backbone	1.0 Hz, 15% IT amplitude	16.27	0.44%	0.006
Scenario 1: Traditional AC-centric	14.8 Hz, 10% IT amplitude	11.39	0.33%	0.236
Scenario 2: AC-fed SST plus 800 VDC baseline	14.8 Hz, 10% IT amplitude	10.90	0.31%	0.214
Scenario 3: Proposed MVDC backbone	14.8 Hz, 10% IT amplitude	10.82	0.29%	0.005

These results should be interpreted as cross-checks of the upstream AC-boundary consequences of the scenario definitions, not as independent site-specific validation. The comparison model alone shows that Scenario 3 has the lowest end-to-end loss. The OpenDSS layer adds that the same scenario also yields the lightest modeled AC-side burden at the campus boundary. In the present scenario, the AC-boundary feeder loss in Scenario 3 is approximately two orders of magnitude lower than in Scenarios 1 and 2 because the long campus-interior AC feeder is removed from the scenario definition. That does not prove a utility interconnection result. It does show that the backbone architecture has a direct and favorable implication at the modeled AC boundary.

OpenDSS harmonic-sensitivity scan

The present paper also adds a standardized OpenDSS harmonic-sensitivity scan at the AC boundary. This is not an IEEE 519 compliance study. Instead, the same three-phase harmonic current probe is injected at the PCC for each scenario at the 5th, 7th, 11th, and 13th harmonic orders, and the resulting harmonic voltage distortion is compared across scenarios. The point of the scan is to test whether the centralized-front-end architecture creates a less sensitive AC boundary under the same perturbation basis.

OpenDSS harmonic-sensitivity scan at the AC boundary

Scenario	Probe THD$_v$ (%)	Max transfer impedance (Ω, L--N)	Interpretation
Scenario 1: Traditional AC-centric	10.02	215.05	Most sensitive AC boundary
Scenario 2: AC-fed SST plus 800 VDC baseline	9.58	205.71	Reduced sensitivity versus Scenario 1
Scenario 3: Proposed MVDC backbone	9.30	199.91	Lowest sensitivity in the present scan

The harmonic-sensitivity result does not establish formal compliance margins, but it does add quantitative support to the power-quality argument. The architecture with the most centralized AC interface also produces the lowest voltage-distortion response under the same harmonic-current probe. That is a stronger basis for the power-quality claim than topology alone, but it remains a comparative sensitivity result rather than a blinded compliance study.

Multi-node campus comparison

Table tab:multinode reports the follow-on campus comparison in which all three scenarios are mapped onto the same four-block topology. This is the strongest topology-level check in the present paper because it removes the objection that Scenario 3 only wins in an overly collapsed equivalent path.

Shared-campus multi-node comparison at 100 MW total IT load

Scenario	Full-load eff.	Source MW	Loss GWh/yr	Worst dynamic source peak (MW)
Scenario 1(M): Traditional AC-centric	81.11%	123.29	204.05	17.86
Scenario 2(M): AC-fed SST plus 800 VDC baseline	85.16%	117.43	152.70	17.03
Scenario 3(M): Multi-node MVDC backbone	86.69%	115.36	134.55	16.81

The shared-campus result is important for two reasons. First, the ranking remains unchanged: the MVDC-backbone case still performs best when the campus is represented as multiple data-center blocks rather than one equivalent path. Second, the gap between Scenario 3(M) and Scenario 2(M) is again modest rather than dramatic, which means the multi-node model supports the architecture direction but does not, by itself, close the case. That is exactly the right interpretation for a position paper rather than a deployment paper. Even so, this shared-campus model is the strongest efficiency layer in the manuscript because it tests the scenarios on the same explicit backbone and block layout.

Public benchmark robustness and dynamic-diversity screens

The manuscript also includes a public-data-only robustness layer to test whether the campus-scale conclusions survive non-local assumptions. In the ESIF-based annualized part-load screen, the single-path result now remains favorable to Scenario 3 relative to the AC-fed SST baseline: Scenario 3 is better than Scenario 2 for 100.0% of annualized hours and yields a smaller total annualized loss. The multi-node comparison is stronger still: Scenario 3(M) is better than Scenario 2(M) for 100.0% of annualized hours and yields a materially smaller annualized loss under the same ESIF bins. The implication is that the architecture case is no longer resting on a perimeter-conversion shortcut, but on a comparison against an explicit SST baseline.

The public RTS-GMLC benchmark screens also make the AC-boundary claim more specific. At the base mirrored-area benchmark location, Scenario 3(M) has a harmonic-sensitivity proxy of 0.00533 pu compared with 0.04088 pu for Scenario 2(M), and a maximum POI voltage-drop proxy of 0.511 percentage points compared with 2.839 for Scenario 2(M). Across the three mirrored RTS areas, the Scenario 2(M)-to-Scenario 3(M) ratio remains at least 4.47$\times$ for the harmonic proxy and 2.85$\times$ for the voltage-drop proxy. These are sensitivity-screen results, not compliance margins, but they are the strongest public-data support in the current paper.

The same direction also survives a common-network benchmark on a public SMART-DS feeder. In that same-feeder screen, Scenario 3(M) produces lower weighted incremental feeder losses, lower burst-driven line-loading change, and more surviving local N--1 cases than Scenario 2(M). When equal total harmonic current is imposed on that same feeder, the maximum POI THD$_v$ proxy remains lower for Scenario 3(M) by 64.49$\times$ to 83.69$\times$ across the tested spectrum families. That result is useful precisely because the total harmonic current is held equal; the remaining separation is therefore an interface-and-network effect rather than a

trivial injection-magnitude effect.

Local N--1 benchmark robustness for Scenario 2(M) and Scenario 3(M)

Area	Valid outages	S2(M) worst incr. load (%)	S3(M) worst incr. load (%)	S2(M) worst $\Delta V_{\text{POI,max}}$	S3(M) worst $\Delta V_{\text{POI,max}}$
Area 1	14	19.26	13.42	5.037	0.511
Area 2	15	17.25	11.63	8.155	2.110
Area 3	14	18.40	12.84	10.381	2.400

Table tab:nminusone extends the same benchmark logic to local one-at-a-time branch outages around each candidate campus area. In the present screen, Scenario 3(M) retains lower incremental loading and lower POI voltage-drop proxy in all three areas. The minimum Scenario 2(M)-to-Scenario 3(M) ratio across the worst local-outage harmonic screens is 6.30$\times$, and the corresponding minimum ratio for the worst local-outage voltage-drop screen is 3.86$\times$. This is still not a post-contingency dispatch or protection study, but it materially strengthens the argument that the upstream AC-boundary advantage is not confined to the intact network.

Dynamic-diversity screen for the multi-node campus model (1.0 Hz, 10% block-level amplitude)

	S2(M) $\Delta P_{\text{src,max}}$ (MW)	S3(M) $\Delta P_{\text{src,max}}$ (MW)	S2(M) $\Delta I_{\text{seg,max}}$ (A)	S3(M) $\Delta I_{\text{seg,m
ampus	11.331	11.204	96.75	161.56
ck only	2.833	2.806	24.18	40.38
e down	0.021	0.016	24.19	40.39
us opposition	0.043	0.032	48.37	80.78
cluster	5.665	5.613	48.37	80.77

Table tab:dynamicdiversity is equally important because it prevents overclaiming. Across all five multi-block dynamic patterns, Scenario 3(M) keeps a slightly lower grid-facing source swing than Scenario 2(M). However, it also produces substantially larger internal segment-current redistribution. The architecture therefore does not reduce every dynamic stress. It shifts some of the burden from the upstream AC boundary into the internal backbone. That tradeoff is a central part of the correct interpretation.

Public RMS dynamic benchmark

The public RMS benchmark strengthens the dynamic case at the upstream AC boundary by moving from quasi-static screens to an electromechanical time-domain comparison on a published dynamic case. Table tab:rmsdynamic shows the 25% campus-share benchmark under the MIT-derived 900 s p95 burst magnitude.

Public RMS dynamic benchmark at 25% campus share and MIT-derived 900 s p95 burst

Grid mode	S2(M) TDS	S3(M) TDS	S2(M) $V_{\text{campus,min}}$	S3(M) $V_{\text{campus,min}}$	S2/S3 max $\|\Delta f\|$ (Hz)
Normal network	OK	OK	0.9222	0.9905	0.602 / 0.574
Remote corridor weakened	OK	OK	0.7118	0.9875	0.678 / 0.584

The RMS result is materially stronger than the earlier quasi-static dynamic layer. Under the weakened-corridor benchmark, Scenario 3(M) improves the minimum campus voltage from 0.7118 pu to 0.9875 pu for the 25% campus-share, 900 s p95 burst case. Across the converged benchmark pairs, the strongest voltage separation reaches 0.2757 pu. Several of the heaviest distributed-interface Scenario 2(M) stress cases do not converge in the published RMS benchmark and are treated here as stability-screen failures rather than as waveform results. This

still does not prove converter dynamics or EMT stability, but it does materially strengthen the claim that centralized AC-boundary exposure is lower in Scenario 3(M) under weak-grid stress.

Reduced-order fault-duty screen

The manuscript now also includes a reduced-order fault-duty screen for the Scenario 3(M) backbone. Table tab:faultscreen reports the baseline 100 mH reactor and 0.01 Ω fault-resistance case at two representative locations.

Reduced-order fault-duty screen for Scenario 3(M)

Location	Prefault current (kA)	5 ms current (kA)	5 ms I^2t (MA^{2}s)	Time to 5 kA (ms)
Remote branch end	0.42	3.81	0.03	6.75
Source-backbone midpoint	1.66	5.10	0.06	4.85

In the full public sweep, the worst 5 ms current occurs at the source-backbone midpoint with a 50 mH reactor and 0.01 Ω fault resistance, reaching 8.53 kA, while the fastest time to 5 kA is 2.43 ms. This is not a protection design. It is a public-data-only interruption-timescale screen. The point is narrower and still important: if the MVDC backbone is taken seriously, millisecond-scale interruption requirements and current-rise behavior become first-order engineering questions rather than deferred implementation details.

Proxy-robustness stress test

Because several advanced-converter curves remain proxy-based, the paper also includes a simple robustness check on the three most consequential proxy curves: the 69 kV AC to 800 VDC perimeter converter, the centralized MV AC/DC front end, and the isolated DC pod stage. In that stress test, each of the three curves is shifted by $\pm$1 percentage point in additive efficiency offset over a five-point grid, yielding 125 combined cases.

The stress test shows that the ranking is not fully robust to proxy uncertainty. Scenario 3 remains more efficient than Scenario 2 in 69 of the 125 single-path cases, and Scenario 3(M) remains more efficient than Scenario 2(M) in 88 of the 125 multi-node cases. This is a useful result precisely because it is not uniformly favorable: it shows that measured front-end and isolated-pod efficiency data are not a cosmetic refinement but a gating requirement for a journal-grade quantitative claim.

Discussion

The results support four main conclusions: a scale-dependent efficiency case for the multi-node backbone, a stronger AC-boundary power-quality case, a stronger AC-boundary voltage-sensitivity case, and a stronger public-benchmark dynamic case for centralized AC-boundary exposure. The evidence for these conclusions is not symmetric, and the paper is stronger when it states that explicitly.

Evidence boundary for the present manuscript

Claim	Strongest current evidence	Present status
Single-path Scenario 3 $>$ Scenario 2 efficiency superiority	Source-anchored path model plus public ESIF annualization	Supported, still proxy-dependent
Multi-node Scenario 3(M) efficiency advantage	Shared-campus model plus public ESIF annualization	Supported across tested scales
Centralized AC-boundary harmonic advantage	Harmonic proxy plus mirrored-area and N--1 screens	Strong at sensitivity-screen level
Centralized AC-boundary voltage advantage	RTS voltage screen plus mirrored-area and N--1 screens	Strong at sensitivity-screen level
RMS weak-grid voltage robustness	Public ANDES benchmark with MIT-derived burst library	Strong at public benchmark level
Protection-timescale plausibility for Scenario 3(M)	Reduced-order RL fault envelope	Partially supported; design not proven

Claim	Strongest current evidence	Present status
Lower internal dynamic stress	Dynamic-diversity screen	Not supported; internal redistribution increases
Deployment-ready MVDC architecture	None	Not proven

Table tab:evidenceboundary summarizes the current evidence boundary. The loss-reduction conclusion is now supported in both the single-path and multi-node comparisons relative to the AC-fed SST baseline, but it remains constrained by proxy uncertainty. The multi-node comparison is still the stronger efficiency layer because the campus topology is explicit and the public ESIF bins keep Scenario 3(M) ahead of Scenario 2(M) across the full tested expansion range.

The power-quality conclusion remains stronger than the efficiency conclusion in the current manuscript. The architecture comparison reduces the number of AC-side interaction points, and the added public harmonic-sensitivity screens show that the centralized-interface Scenario 3(M) remains materially less sensitive than Scenario 2(M) across mirrored benchmark areas and local one-at-a-time branch outages. This still does not replace a full IEEE 519 study with converter-specific spectra and utility short-circuit data. It does justify a narrower claim: centralizing the AC interface appears favorable for grid-facing harmonic sensitivity under the public benchmark assumptions used here.

The upstream voltage-sensitivity conclusion is similarly strong at the benchmark-screen level. Scenario 3(M) consistently shows lower POI voltage-drop proxy than Scenario 2(M) at the base mirrored benchmark locations and in the local N--1 screens. This is again a sensitivity-screen result rather than a site-specific interconnection result, but it is exactly the kind of architecture-level evidence that a white paper should present clearly.

The public RMS benchmark sharpens that same conclusion under a stronger dynamic test. On the published IEEE 14 dynamic case with MIT-derived burst magnitudes, Scenario 3(M) keeps materially higher minimum campus voltage than Scenario 2(M), especially when the remote corridor is weakened before the burst step. This is still not converter-EMT evidence, but it is much closer to a true disturbance-response screen than the earlier quasi-static dynamic layer.

The dynamic-diversity screen changes the interpretation of the integration argument. The present model supports the claim that Scenario 3(M) slightly reduces grid-facing source swing across coherent and non-coherent block patterns. It does \emph{not} support a blanket claim that the architecture reduces all dynamic stress. Internal backbone current redistribution increases materially in Scenario 3(M), which means the benefit is primarily at the upstream AC boundary and not an automatic reduction of all internal transients.

The new fault-duty screen sharpens the protection discussion in a different way. It does not prove a selective protection design, but it does bound the order of magnitude of fault-current rise and the interruption timescale that the backbone would force onto any practical breaker concept. That means protection is no longer a vague future-work placeholder in the manuscript; it is now a bounded unresolved problem with explicit millisecond timescale implications.

The paper should therefore defend six bounded statements:

- The single-path Scenario 3 result is favorable relative to the AC-fed SST baseline, but it remains quantitatively modest and proxy-dependent.
- The multi-node Scenario 3(M) result is the stronger efficiency case and remains favorable across the tested campus-growth range.

- Centralizing the AC boundary materially improves the present public-network harmonic and voltage sensitivity screens.
- The public RMS benchmark strengthens the case that Scenario 3(M) improves upstream weak-grid voltage robustness under burst loading.
- Dynamic evaluation is mandatory, and the present evidence shows both an upstream benefit and an internal-backbone tradeoff.
- The fault-duty screen bounds interruption timescales, but deployable MVDC protection remains an open design problem.

The paper should not claim that the present model proves exact harmonic compliance, exact converter-control stability margins, deployable MVDC protection settings, or universal superiority of Scenario 3 over Scenario 2 under all operating assumptions.

Research Priorities

The present results establish a defensible architecture-level case for an MVDC backbone, but several research questions remain open before the concept can be treated as deployment-ready. The most important next steps are the following.

- Protection and grounding: Build from the present reduced-order fault envelope into equipment-specific fault-clearing, grounding, insulation-coordination, and sectionalization concepts for the MVDC backbone.
- Converter characterization: Replace proxy efficiency curves with vendor-qualified or experimentally measured curves for the centralized rectifier and isolated DC pod stages.
- PCC studies: Extend the present AC-boundary model into a site-specific interconnection study using utility Thevenin equivalents, harmonic limits, and short-circuit data.
- Dynamic controls: Build from the public RMS benchmark into converter-aware EMT studies with explicit control models for the centralized front end, isolated DC pod, and supporting buffers.
- Buffer placement: Determine the preferred placement and control role of subsecond energy buffers across the GPU, rack, pod, and backbone levels.

Among these topics, the highest-value next modeling step is an EMT-grade validation of the centralized front end, isolated DC pod, and buffer controls under weak-grid and disturbance scenarios. That step would materially strengthen the case that the architecture remains favorable not only in steady-state comparisons, but also under converter-driven transient conditions that matter for large AI-factory interconnections.

Conclusion

This paper argues that the next logical move after the 800 VDC facility bus is to seriously evaluate a DC subtransmission backbone. The position is motivated by three facts that now occur together: AI factories are becoming very large loads, they are increasingly dynamic loads, and the dominant controllable resources around them are already DC-native.

The present source-anchored and public-benchmark results support that position more strongly once Scenario 2 is treated as an explicit AC-fed SST plus 800 VDC baseline. At the 100 MW reference campus, Scenario 3, the

proposed MVDC backbone, remains the best case in the single-path reference profile, although its margin over Scenario 2 is still modest. The stronger result comes from the shared-campus comparison and the public benchmark screens. When all three scenarios are forced onto the same explicit four-block campus topology, Scenario 3(M) still yields the best efficiency. More importantly, it also yields the strongest public-data case for lower AC-boundary harmonic sensitivity and lower AC-boundary voltage sensitivity, and those advantages persist in mirrored benchmark areas and local one-at-a-time branch-outage screens.

The dynamic evidence also becomes more precise in this framing. The architecture question does not disappear when AI loads are treated as time-varying, but the benefit is not uniform. Scenario 3(M) slightly lowers grid-facing source swing in the present dynamic-diversity screen, while also increasing internal backbone current redistribution. The public RMS benchmark strengthens the upstream side of that argument by showing materially higher minimum campus voltage for Scenario 3(M) under MIT-derived burst steps, especially when the remote AC corridor is weakened. The correct conclusion is therefore not that the backbone eliminates dynamic stress. It is that it appears to shift dynamic stress away from the upstream AC boundary and into an internal DC domain that still requires dedicated control and protection design.

The protection evidence is also sharper than before. The reduced-order fault screen does not deliver breaker settings or a protection philosophy, but it does show that representative source-side backbone faults can reach the 5 kA level within a few milliseconds under public benchmark assumptions. That makes protection and interruption timescale a first-order design topic rather than a vague future concern.

The result is not a claim that every deployment question is finished. It is a claim that the right campus-scale reference architecture has become visible. For AI factories, the higher-value move is no longer only a better facility front end. It is a campus architecture that treats the DC nature of the load as a first-order design principle while being explicit about the limits of the current evidence.

Reproducibility Note

The quantitative results in this manuscript are generated from the project files dc_backbone_model.py, scientific_assumptions_v1.json, source_backed_model_report.json, dc_backbone_multinode_campus_model.py, and multinode_campus_report.json. The single-path OpenDSS validation can be reproduced with:

```
python3 dc_backbone_model.py --run-opendss \
  --save-json source_backed_model_report.json \
  --write-memo MODEL_RESULTS_MEMO.md
```

The multi-node campus comparison can be reproduced with:

```
python3 dc_backbone_multinode_campus_model.py \
  --save-json multinode_campus_report.json
```

The public benchmark and robustness screens can be reproduced with:

```
python3 dc_backbone_public_benchmark_model.py \
  --save-json public_benchmark_report.json

python3 dc_backbone_public_benefit_analysis.py \
  --save-json public_benefit_report.json \
  --save-note PUBLIC_BENEFIT_ANALYSIS.md
```

The public operating, common-network, RMS, and fault screens can be reproduced with:

```
python3 public_time_series_ai_factory.py --details

python3 public_common_network_td_study.py
```

```
python3 public_harmonic_frequency_sweep.py --details

python3 public_rms_dynamic_study.py

python3 public_fault_envelope.py --details
```

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